

## PORT ORFORD, OR, USA

WMO: 994031

Lat: 42.7390N

Lon: 124.4980W

Elev: 8

StdP: 101.23

Time Zone: -8.00 (NAP)

Period: 05-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 2.3        | 3.5 | N/A                           | N/A | N/A  | N/A | N/A | N/A  | 18.0                  | 10.7 | 16.2 | 11.3 | 2.7                   | 0    | 0.653 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|-----|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%  |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                  | (p)  |
| 8             | 5.2                    | 21.5            | N/A  | 20.1 | N/A  | 19.1 | N/A  | N/A                 | N/A  | N/A | N/A  | N/A | N/A  | 6.2                  | 340  |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| N/A                             | N/A | N/A  | N/A | N/A | N/A  | N/A | N/A | N/A  | N/A           | N/A  | N/A  | N/A  | N/A  | N/A  | N/A               |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 16.0          | 13.8 | 11.6 | DB                         | -0.1 | 26.0               | 1.7 | 2.8                                                | -1.3 | 28.0       | -2.3 | 29.7       | -3.2 | 31.3       | -4.4 | 33.4 | (4) |
| (5)               |      |      | WB                         | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  | N/A  | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 11.6   | 9.4   | 8.9   | 9.0   | 9.6   | 11.7  | 13.4  | 14.9  | 15.2  | 14.5  | 13.0  | 10.7  | 9.1  |
|                                                                                  | DBStd       | 3.22   | 2.44  | 2.54  | 2.03  | 2.02  | 2.06  | 1.69  | 1.86  | 1.94  | 2.23  | 2.13  | 2.25  | 2.71 |
|                                                                                  | HDD10.0     | 233    | 40    | 46    | 44    | 29    | 4     | 1     | 1     | 0     | 1     | 2     | 17    | 49   |
|                                                                                  | HDD18.3     | 2448   | 277   | 263   | 290   | 263   | 208   | 149   | 106   | 97    | 115   | 167   | 228   | 285  |
|                                                                                  | CDD10.0     | 832    | 22    | 16    | 12    | 17    | 56    | 101   | 153   | 162   | 137   | 94    | 39    | 22   |
|                                                                                  | CDD18.3     | 5      | 0     | 0     | 0     | 1     | 1     | 1     | 0     | 1     | 1     | 1     | 0     | 0    |
|                                                                                  | CDH23.3     | 18     | 0     | 0     | 0     | 2     | 8     | 1     | 1     | 1     | 3     | 3     | 0     | 0    |
|                                                                                  | CDH26.7     | 2      | 0     | 0     | 0     | 0     | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 4.9    | 5.1   | 4.7   | 5.7   | 4.7   | 4.7   | 5.0   | 5.2   | 4.6   | 4.1   | 4.4   | 4.8   | 5.6  |
| Precipitation                                                                    | PrecAvg     | 2079   | 351   | 248   | 242   | 192   | 108   | 57    | 9     | 19    | 33    | 134   | 292   | 382  |
|                                                                                  | PrecMax     | 3334   | 651   | 520   | 406   | 439   | 240   | 192   | 53    | 95    | 166   | 286   | 636   | 756  |
|                                                                                  | PrecMin     | 1529   | 136   | 97    | 77    | 74    | 0     | 4     | 0     | 0     | 0     | 5     | 78    | 136  |
|                                                                                  | PrecStd     | 450    | 152   | 125   | 95    | 100   | 62    | 43    | 13    | 25    | 37    | 74    | 142   | 149  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 17.2  | 17.8  | 15.7  | 21.7  | 23.5  | 21.2  | 22.3  | 21.7  | 22.4  | 22.6  | 18.5  | 16.4 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 2%          | DB     | 15.0  | 14.9  | 13.9  | 15.6  | 18.2  | 18.7  | 20.1  | 20.5  | 20.5  | 19.1  | 15.9  | 15.0 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 5%          | DB     | 13.7  | 13.5  | 12.8  | 13.4  | 15.8  | 17.2  | 19.1  | 19.5  | 19.1  | 17.1  | 14.7  | 13.9 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 10%         | DB     | 12.6  | 12.5  | 11.8  | 12.1  | 14.7  | 16.2  | 18.3  | 18.6  | 17.9  | 15.8  | 13.9  | 12.9 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 4.5   | 4.7   | 4.4   | 4.6   | 4.8   | 4.8   | 5.3   | 5.2   | 5.3   | 4.8   | 4.5   | 4.6  |
|                                                                                  |             | MCDBR  | 6.2   | 6.3   | 6.0   | 7.2   | 7.4   | 6.9   | 6.7   | 6.7   | 7.6   | 7.3   | 5.8   | 5.1  |
|                                                                                  |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.320  | 0.322 | 0.329 | 0.356 | 0.361 | 0.351 | 0.345 | 0.348 | 0.339 | 0.333 | 0.320 | 0.322 |      |
|                                                                                  | taud        | 2.563  | 2.564 | 2.563 | 2.445 | 2.473 | 2.511 | 2.535 | 2.547 | 2.547 | 2.577 | 2.585 | 2.555 |      |
|                                                                                  | Ebn at Noon | 827    | 886   | 920   | 913   | 912   | 919   | 921   | 910   | 896   | 861   | 820   | 789   |      |
|                                                                                  | Edh at Noon | 69     | 79    | 89    | 108   | 108   | 104   | 101   | 96    | 90    | 77    | 66    | 64    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.61   | 2.41  | 3.54  | 5.02  | 6.34  | 6.98  | 7.31  | 6.15  | 4.88  | 3.09  | 1.80  | 1.31  |      |
|                                                                                  | RadStd      | 0.24   | 0.31  | 0.39  | 0.41  | 0.36  | 0.42  | 0.29  | 0.37  | 0.28  | 0.29  | 0.22  | 0.24  |      |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (12 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | +2      |

Nomenclature: See separate page